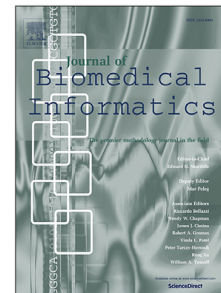


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Personalized Federated Learning for Medical Vision-Language Models via Efficient Fine-Tuning and Uncertainty-Aware Disentanglement

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Abstract

Privacy concerns and data heterogeneity remain critical barriers to advancing medical AI. While federated learning (FL) offers a decentralized solution, it often suffers from the *Subspace Mismatch* problem in clinical settings, where non-independent and identically distributed (non-IID) data drives clients towards divergent parameter manifolds. To address this, we propose a novel personalized federated learning (pFL) framework for vision-language medical tasks. Our approach employs a “separate-and-calibrate” strategy: First, we leverage Parameter-Efficient Fine-Tuning (PEFT) with a *Dual-Subspace Disentanglement* mechanism, splitting parameters into shared and personalized components to decouple general medical knowledge from site-specific characteristics. Second, to prevent negative transfer, we introduce Dynamic Likelihood-weighted Uncertainty Calibration (DLUC). This mechanism utilizes Dempster-Shafer theory to quantify epistemic uncertainty, acting as a reliability-aware gate that selectively aggregates compatible knowledge. Extensive experiments on Visual Question Answering, Question Generation, and Medical Report Generation demonstrate that our method significantly mitigates client drift and achieves superior personalization compared to state-of-the-art baselines.

Keywords: Personalized federated learning, large vision-language medical model, dempster-shafer theory

1. Introduction

Privacy protection remains a significant challenge in the application of deep learning to the medical field, primarily due to stringent regulations such as the Health Insurance Portability and Accountability Act [1] and the General Data Protection Regulation [2]. Furthermore, medical data is highly heterogeneous and non-independent and identically distributed (non-IID). This complexity arises from anatomical differences and regional variations across institutions, adding challenges to the development of medical AI models [3]. To address these challenges, federated learning (FL) [4] presents a promising solution by enabling decentralized model training through parameter sharing, eliminating the need for data centralization. In this approach, individual clients train local models on protected data, and the aggregated parameters are used to update a global model, which captures shared knowledge across all clients while preserving data privacy. In recent years, researchers have explored the potential of FL for various medical tasks, with a focus on developing a robust global model that integrates knowledge across multiple clients [5, 6]. Building upon FL, personalized federated learning (pFL) [7] is specifically designed to enhance the performance of individual clients rather than focusing on a central global model. This approach aligns well with the demands of the medical sector but reveals new opportunities for improving client-specific information aggregation within the FL framework.

Most current federated medical studies have centered on simple tasks with structured outputs, such as segmentation [8] and

classification [9] [10] [11]. Multimodal medical tasks [12], which involve the integration of both vision and language modalities, remain largely unexplored within the pFL framework. In particular, vision-language tasks like medical Visual Question Answering (VQA) [13], Visual Question Generation (VQG) [14], and medical report generation (MRG) [15] have the potential to significantly improve human interpretability and clinical decision support. However, existing pFL methods have yet to fully address the unique challenges of vision-language tasks in the medical domain.

The development of FL for medical vision-language tasks faces two primary hurdles: computational demands and the *Subspace Mismatch* problem derived from client drift. While Large Language Models (LLMs) and Vision-Language Models (VLMs) [16] offer powerful capabilities, their massive parameter scales result in prohibitively high computational and communication costs in pFL, rendering direct parameter sharing impractical. More critically, simply reducing communication via Parameter-Efficient Fine-Tuning (PEFT) introduces a subtle but severe issue: PEFT modules optimize within low-rank manifolds. Under non-IID medical data distributions, clients optimize towards divergent parameter subspaces. Naively aggregating these mismatched subspaces leads to negative transfer, where the shared knowledge conflicts with the local feature representation [17]. In medical research, where personalization is paramount, global regularization strategies [18] often prove counterproductive.

To simultaneously tackle computational efficiency and subspace alignment, we propose a novel separate-and-calibrate

strategy. First, we integrate PEFT methods—specifically Prompt-tuning [19], Prefix-tuning [20], and Low-Rank Adaptation (LoRA) [21]—decomposed into global (shared) and personalized (local) components. This ensures that while general knowledge is circulated, client-specific feature extraction remains undisturbed. Crucially, to prevent negative transfer from mismatched subspaces, we introduce Dynamic Likelihood-weighted Uncertainty Calibration (DLUC). Instead of treating all peer updates as equally valid, DLUC utilizes Dempster-Shafer theory (DST) [22] to quantify the *epistemic uncertainty* [23] of incoming parameters. This mechanism acts as a distribution-aware gate: if a peer’s parameters induce high uncertainty on local data (indicating a domain shift), their contribution is down-weighted [24]. This Uncertainty-Guided Subspace Alignment ensures that clients only assimilate compatible knowledge.

In summary, the main contributions of this paper are:

- **Framework Addressing Subspace Mismatch:** We propose a novel Dual-Subspace Disentanglement personalized Federated Learning (DSD-pFL) framework tailored for medical VLMs. By differentiating between shared global knowledge and client-specific parametrization within the PEFT manifold, we effectively mitigate the subspace mismatch problem common in non-IID clinical settings.
- **Epistemic Uncertainty-Guided Aggregation:** We introduce DLUC, an evidence-driven calibration mechanism. By leveraging DST to quantify epistemic uncertainty as a proxy for distribution shift, DLUC enables clients to selectively assimilate high-confidence knowledge while filtering out out-of-distribution (OOD) noise.
- **Validation on Complex Multimodal Tasks:** We empirically validate our approach across Visual Question Answering (VQA), Visual Question Generation (VQG), and Medical Report Generation (MRG). Results demonstrate that our method achieves superior personalization and robustness compared to state-of-the-art baselines.

2. Related Work

2.1. Federated Learning in Medical Applications

Medical data often face challenges of limited sample sizes and significant heterogeneity, compounded by privacy regulations [1, 2, 25] that prevent addressing these issues through data sharing. FL enables the collaborative training of machine learning models across different sites without the need for data exchange or sharing. It presents a promising solution for mitigating small sample size challenges [26] while safeguarding individual privacy. Although FL has been applied across various medical domains—including diabetic retinopathy [27, 28], medical image classification [29], cancer diagnosis [30, 31, 32], pneumonia recognition [33], and COVID-19 detection [34]—its application to multimodal scenarios remains underexplored, particularly in handling modality alignment and client drift in

medical settings. Standard FL struggles with fusing heterogeneous data like images and text across institutions with varying protocols. To address this, we introduce a framework that leverages PEFT and uncertainty-aware aggregation, enhancing both personalization and cross-client knowledge sharing in multimodal medical AI.

2.2. Medical Personalized Federated Learning

pFL aims to adapt global models to individual clients, addressing the unique needs of each client while benefiting from collaborative training. While computational burden and client drift are general challenges for all FL methods, most existing pFL approaches have substantially reduced communication overhead by sharing only a small subset of parameters [35, 36, 37]. The challenge unique to pFL—particularly in the medical domain—is effectively balancing personalization and generalization: enabling each client to capture institution-specific clinical patterns while benefiting from globally shared knowledge across heterogeneous data distributions.

Computational Burden is a recognized bottleneck in FL that pFL methods must also address when scaling to large VLMs. The computational and communication costs scale with the model size and the number of participating clients, hindering scalability and efficiency [38]. PEFT methods allow models to be tuned with fewer parameters, reducing computational costs while maintaining performance. Recent advancements attempt to merge PEFT with FL to reduce overhead. Zhu et al. [39] utilized prompt learning for uni-modal FL, but their approach lacks valid personalization mechanisms for complex multimodal tasks. Similarly, theories like Meta-Learning (e.g., MAML) share conceptual similarities with pFL but often require substantial local computation. Additionally, recent studies have explored semi-asynchronous strategies [40] and cluster-based personalization [41] to improve federated medical image analysis. However, these methods are less feasible for resource-constrained medical edge devices compared to lightweight PEFT. Our work bridges this gap by extending PEFT into a dual-subspace pFL framework specifically optimized for the high-dimensional manifolds of medical VLMs.

Client Drift arises due to substantial heterogeneity in medical image distributions, leading to divergence in local model optimizations during federated training [17]. Existing methods to mitigate client drift generally fall into two categories: model splitting and client selection. The model splitting approach divides the model into global and personalized components, where the global component learns shared features across clients, and the personalized component is fine-tuned locally [35, 36, 37]. This approach aims to balance shared learning with client-specific adaptation, but it may still be insufficient in highly heterogeneous data settings. Client selection, on the other hand, seeks to improve federated model aggregation by dynamically determining which clients’ updates are most beneficial. Zhang et al. [42] proposed a dynamically fused FL approach for COVID-19 diagnosis, where client selection was based on local model performance, enabling more effective aggregation of relevant knowledge. Additionally, Qin et al. [24]

introduced RIPFL, which integrates social and personal information using the Dempster-Shafer evidence theory to improve client selection and reduce the negative impact of client drift. However, the application of such strategies in multimodal medical tasks remains limited.

Statement of Significance:

Problem	Existing pFL methods are ill-suited for multimodal medical tasks due to high computational demands and the subspace mismatch caused by clinical data heterogeneity.
What is already known	FL enables privacy-preserving collaboration, but standard approaches struggle to align diverse medical data distributions, particularly for complex vision-language models.
What this paper adds	We introduce a reliable pFL framework that disentangles global/local knowledge via partitioned PEFT and employs a Dempster-Shafer theory-based uncertainty gate to robustly aggregate compatible representations.
Who would benefit from the new knowledge	Medical AI researchers and healthcare institutions seeking practical, privacy-compliant solutions for deploying sophisticated multimodal assistants across hospital networks.

3. Methodology

3.1. Problem Formulation

An overview of our method is depicted in Fig. 1. We consider a pFL setting comprising T separate clients, denoted as $\{C_t\}_{t=1}^T$, where each client is initialized with a pre-trained model and fine-tuned on local distributed datasets $\{D_t\}_{t=1}^T$ via PEFT. For a client model C_t parameterized by θ_t , we formulate the local optimization as a joint maximization problem: $\arg \max_{\theta_t} Q(G, U|\theta_t, D_t)$. Here, Q represents the rigorous quality assessment of the model's output. Uniquely, our objective function does not solely maximize the likelihood of the generated content G (accuracy); it simultaneously calibrates the estimated uncertainty U (reliability). This ensures that the local model learns to yield high confidence only for in-distribution data, providing a necessary prerequisite for the subsequent trust-based aggregation.

To achieve this, we adopt a **Server-Client Federated Architecture**. Unlike Peer-to-Peer (P2P) systems where clients communicate directly, our framework employs a central server as a secure aggregation hub. The objective of the inter-client communication, mediated by the server, is to maximize the transfer of useful knowledge:

$$\arg \max_{\mathbb{E}} \sum_{t=1}^T Q(G|\mathbb{E}, D_t, \theta_1, \dots, \theta_T), \quad (1)$$

where $\mathbb{E}$ represents the DLUC-driven aggregation policy. By optimizing these dual objectives, we ensure that clients not only learn from their own data but also selectively absorb high-fidelity knowledge from the server, effectively bridging general medical insights with local clinical specifics.

3.2. Local Training for Client Vision-language Medical Model

This section describes the local training process. For client C_t , the dataset D_t comprises n instances that consist of a vision-language input pair $\{V_t, L_t\}$, and the corresponding

ground truth output. We utilize transformer-based encoders E_V and E_L to extract the visual features and the language features from V_t and L_t separately. These extracted features are then fused into a unified input embedding e_t , where $e_t \in \mathbb{R}^{b \times m}$ (b is the input sequence length and m is the feature dimension). The fused embedding e_t is subsequently processed by a decoder M_t , which generates the predicted outputs G_t .

3.2.1. Parameter-efficient Fine-tuning in Client Model

In the proposed method, we introduce three PEFT mechanisms separately: prompt-tuning, prefix-tuning, and LoRA. Crucially, to effectively decouple generalizable medical knowledge from site-specific variations, we propose a *Dual-Subspace Disentanglement* strategy. We partition the PEFT parameters into two distinct sets: a **Shared (Global) Component** and a **Private (Local) Component**. The Shared Component is aggregated across clients to learn universal representations, such as anatomical structures and general medical syntax, benefiting from the large-scale federated data. The Private Component is kept strictly local, allowing it to adapt to the unique "client-specific subspace"—encompassing factors like specific MRI scanner artifacts and hospital-specific reporting styles—without interference from external distributions. This explicit separation prevents the contamination of global consensus by local biases, mitigating the subspace mismatch problem described above.

Prompt-tuning assigns each client a set of trainable prompt embeddings, which are added to the model's input sequence, while the pre-trained model weights remain frozen. We add prompts P to the input embeddings e_t . The prompts are divided into local and shared components, denoted as $P = \{P_l, P_s\}$; $P_l \in \mathbb{R}^{d \times m}$, $P_s \in \mathbb{R}^{d \times m}$ (d is the length of the prompt). The final decoding process can be calculated as follows:

$$G_t = D([P_l, P_s, e_t]). \quad (2)$$

Prefix-tuning learns a set of trainable prefix embeddings $\mathbf{p} \in \mathbb{R}^{l \times m}$ (l is the total prefix length, m is the hidden dimension of the transformer) for each client, which is directly injected into the multi-head attention (MHA) layers of the encoders E_V and E_L . $\mathbf{p}$ is also composed of local and shared components, which can be represented as $\mathbf{p} = \{\mathbf{p}_l, \mathbf{p}_s\} \in \mathbb{R}^{\frac{l}{2} \times m}$. These prefixes serve as a context for influencing the self-attention computation. Specifically, we split $\mathbf{p}$ into $\{\mathbf{p}^K, \mathbf{p}^V\} \in \mathbb{R}^{l/2 \times m}$, which are composed of half of $\mathbf{p}_l$ and $\mathbf{p}_s$, both with the same dimensions $\frac{l}{4} \times m$. The prefixes are introduced into the MHA layer as follows:

$$\text{Attention}(\mathbf{Q}, \hat{\mathbf{K}}, \hat{\mathbf{V}}) = \text{Softmax}\left(\frac{\mathbf{Q}\hat{\mathbf{K}}^T}{\sqrt{m}}\right)\hat{\mathbf{V}}; \\ \hat{\mathbf{K}} = [\mathbf{p}^K; \mathbf{K}], \hat{\mathbf{V}} = [\mathbf{p}^V; \mathbf{V}], \quad (3)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the original inputs of the MHA layer. **LoRA** applies low-rank decomposition to the weight changes in the pre-trained model, reducing the number of fine-tuning parameters by adding a low-rank trainable module. We define local and shared LoRA parameters, denoted as $\Delta\theta_l = \mathbf{B}_l\mathbf{A}_l$ and

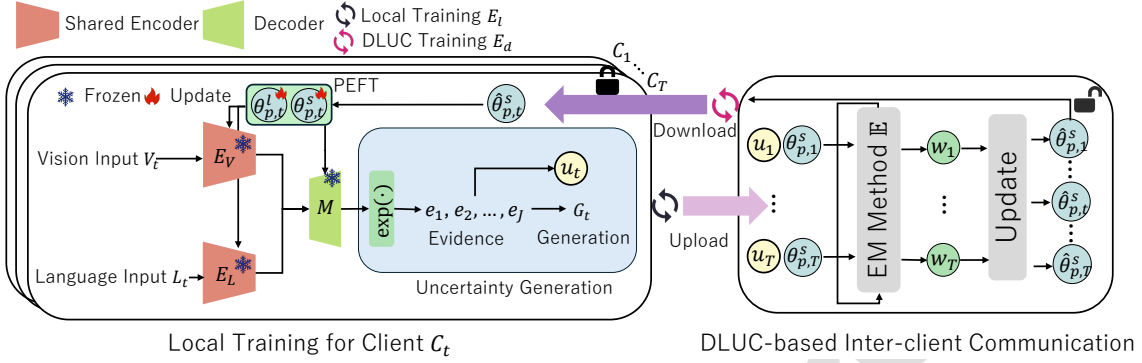


Figure 1: Overview of the proposed pFL approach for the vision-language medical task. The local client training integrates PEFT-based prompt tuning, where vision and language encoders are partially frozen. Uncertainty generation aids in dynamic inter-client prompt exchange via the EM method, optimizing personalization and adaptation in federated learning.

$\Delta\theta_s = \mathbf{B}_s\mathbf{A}_s$, respectively. $\mathbf{A}$ and $\mathbf{B}$ are low-rank matrices, and $\Delta\theta$ represents the fine-tuning changes to the model parameters θ_t . The new model weights can be represented as:

$$\theta_t = \theta_t + \Delta\theta_l + \Delta\theta_s$$

Since the parameters updated by each method are different, we uniformly define the updated parameters as $\theta_p = \{\theta_p^l, \theta_p^s\}$, where θ_p^l and θ_p^s represent the local and shared parameters, respectively. Reducing the number of parameters involved in inter-client communication through PEFT methods can effectively address the issue of computational burden.

3.2.2. Dempster-Shafer Theory based Prediction

Conventional deep learning classifiers typically utilize a Softmax activation function to output probability distributions. However, Softmax is known to produce overconfident predictions even for out-of-distribution (OOD) inputs, as it forces the output vector to sum to 1 regardless of the evidence strength. This limitation makes it unsuitable for detecting subspace mismatch in FL. To rigorously quantify the model's ignorance (Epistemic Uncertainty), we adopt DST to model the output as a Dirichlet distribution parameterized by evidence counts $\alpha_j = \{\alpha_j\}_{j=1}^J$. Specifically, we define the output hidden states of the decoder M as the evidence $\mathbf{e} = \{e_j\}_{j=1}^J$ of each candidate's word. Because evidence must be non-negative, we replace the final softmax layer of the conventional model with an $\exp(\cdot)$ activation layer, yielding evidence $e_j = \exp(f(D_t|\theta_t))$. The Dirichlet parameter $\alpha = \{\alpha_j\}_{j=1}^J$ can be calculated by evidence as $\alpha_j = e_j + 1$. Based on α , the belief mass $\mathbf{b} = \{b_j\}_{j=1}^J$ of the model for each candidate's word can be defined as $b_j = \frac{e_j}{S}$, where $S = \sum_{j=1}^J \alpha_j$. Unlike traditional probability distributions, the sum of $\mathbf{b}$ does not equal 1, and the difference from 1 is the uncertainty u of the model, which can be calculated as $u = 1 - \sum_{j=1}^J b_j$; $u > 0, b_j > 0$.

To assign more evidence to the correct word, we employ the

following Dirichlet term:

$$\begin{aligned} \text{Dirichlet}(\theta_t) &= \int \frac{\Gamma(\alpha_1 + \dots + \alpha_j)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_j)} \prod_{j=1}^J p_j^{\alpha_j-1} d\mathbf{p}_j \\ &= \int \frac{1}{B(\alpha_1, \dots, \alpha_j)} \prod_{j=1}^J p_j^{\alpha_j-1} d\mathbf{p}_j \\ &= \int \frac{1}{B(\alpha_j)} \prod_{j=1}^J p_j^{\alpha_j-1} d\mathbf{p}_j, \end{aligned} \quad (4)$$

where $B(\cdot)$ is the beta function. Here, $\mathbf{p}_j$ represents the class probabilities on the simplex. The Dirichlet term is introduced into the cross-entropy loss function according to obtain $\mathcal{L}^{\text{unc}}(\cdot)$ as follows:

$$\mathcal{L}^{\text{unc}}(\theta_t) = \sum_{j=1}^J y_j(\psi(S_j) - \psi(\alpha_j)), \quad (5)$$

where y_j is the one-hot vector of the ground truth word, and $\psi(\cdot)$ is the digamma function. Note that Eq. (5) aims to allocate the sum of all evidence generated by predictions as much as possible to the correct answers, providing positive feedback. However, since the above loss does not ensure that incorrect labels yield less evidence, the KL term is introduced to minimize the evidence for incorrect labels as much as possible to 0:

$$\begin{aligned} \text{KL}[D(\mathbf{p}_j|\hat{\alpha}_j)||D(\mathbf{p}_j|\mathbf{1})] &= \log\left(\frac{\Gamma(\sum_{j=1}^J \hat{\alpha}_j)}{\Gamma(J) \prod_{j=1}^J \Gamma(\hat{\alpha}_j)}\right) \\ &+ \sum_{j=1}^J (\hat{\alpha}_j - 1)[\log(\hat{\alpha}_j) - \log(\sum_{i=1}^J \hat{\alpha}_i)]. \end{aligned} \quad (6)$$

Here, $\mathbf{1}$ is a parameter vector of K ones, and $\hat{\alpha}_j = \mathbf{y}_j + (1 - \mathbf{y}_j) \odot \alpha_j$ is the Dirichlet parameter to prevent penalties for correct evidence (when $y_j = 1, \hat{\alpha}_j = 1$ and the loss becomes 0). Therefore, the final loss function can be calculated as follows:

$$\mathcal{L}(\theta) = \mathcal{L}^{\text{unc}}(\theta) + \lambda \text{KL}[D(\mathbf{p}_j|\hat{\alpha}_j)||D(\mathbf{p}_j|\mathbf{1})], \quad (7)$$

Algorithm 1 DSD-pFL: Dual-Subspace Disentanglement Personalized Federated Learning for Vision-Language Medical Tasks

Require: E_g -Number of global epoch, E_L -Number of local epoch, α - learning rate, T -Number of clients
 1: Initialization: client set $C = \{p_t^p, p_t^s, w_t^{EM}\}_{t=1}^T$
 2: **for** $e_g = 1$ to E_g **do**
 3: **for** $t = 1$ to T **do**
 4: **for** $e_l = 1$ to E_L **do**
 5: $\theta_{p,t}^s = \theta_{p,t}^s - \alpha \nabla_{\theta_{p,t}^s}(\mathcal{L})$
 6: {Uncertainty μ_t is computed internally}
 7: **end for**
 8: $\eta = \text{DLUC}(\{\mu_t\}_{t=1}^T, \theta_t)$
 9: $\theta_{p,t}^s = \sum_{i=1}^T \eta_i \theta_{p,i}^s$
 10: **end for**
 11: **Ensure:** $C = \{p_t^p, p_t^s, w_t^{EM}\}_{t=1}^T$

Algorithm 2 DLUC Process

Require: E_d -Number of DLUC Epoch, β -learning rate
 $\{\mu_t\}_{t=1}^T$ -clients' uncertainty list, w_t^{EM} - $\mathbb{E}$'s parampter
 1: **for** $e_d = 1$ to E_d **do**
 2: $w_t^{EM} = w_t^{EM} - \beta(\nabla_{w_t^{EM}} \mathcal{L}_{\text{client}} + \nabla_{w_t^{EM}} \|w_t^{EM}\|_2^2)$ {Update EM network weights to minimize client aggregation loss with L2 regularization}
 3: **end for**
 4: $\eta = \mathbb{E}(\{\mu_t\}_{t=1}^T, w_t^{EM})$ {Compute aggregation weights from client uncertainties via the learned EM network}
 5: **return** $\eta = \{\eta_1, \dots, \eta_T\}$

where λ is a balance parameter. The client model can output each word and the uncertainty U_n by training the model using the proposed loss function. In the inter-client communication process, uncertainty evaluates the model's performance.

3.3. DLUC-based Inter-client Communication Process

The core objective of DLUC is to construct a *Reliability-Aware Aggregation Gate*. Instead of blindly averaging parameters (which leads to negative transfer), we define a dynamic weight w_t serving as a "trust score" for each peer. This weight determines the proportion of the peer's shared prompt $\theta_{p,t}^s$ that effectively aligns with the local subspace. Given the data D_t and the clients' parameters θ_t , the conditional expectation, as a lower bound on the true likelihood, is computed as

$$\mathbb{E}[\log \Pr[D_t | \theta_{p,t}]] = \sum_{i=1}^T \log \Pr(A_t | D_t, \theta_t, \theta_{p,t}^s + \mu \theta_{p,t}^s), \quad (8)$$

where $\theta_{p,t}^s = \sum_{k=1}^{T, k \neq t} w_k \theta_{p,k}$, μ is an aggregation rate, and the w_t is the weight optimized by the EM method $\mathbb{E}$. Since the prompts represent the client data schema, we use a neural network to optimize the objective of Eq. (8). We take the uncertainty u as the network input and the weights w as the output. During training, we set up local training step E_l and

inter-client communication E_c separately, representing the number of iterations each is executed, and cross these two processes. This DLUC aims to find the best configuration of weights to maximize the total likelihood of the model for all client data. This optimization allows the client to aggregate the maximum amount of information from the server while avoiding the interference of invalid information. The main procedure is summarized in Algorithms 1 and 2. In Algorithm 2, step 2 updates the aggregation weights via gradient descent to minimize the local objective along with an L_2 regularization term, ensuring a smooth weight distribution.

4. Experiments

4.1. Datasets

This study is driven by the significant impact of transformer-based architectures on multimodal medical tasks. Considering the wide range of applications for VQA and VQG tasks, we selected medical images and QA pairs from the VQA-RAD [43] and Slake [44] datasets. These datasets were further partitioned into sub-datasets based on the anatomical focus of the images. Reflecting the structure of different departments in a real clinical setting, we defined four clients for Slake, referred to as Client 1–4, which include "Lung" (734 question pairs), "Abdomen" (829 question pairs), "Brain" (461 question pairs), and "Others" (including "Chest", "Neck", and "Pelvic" with 335 question pairs). Similarly, for VQA-RAD, we set three clients, referred to as Client 1–3: "Chest" (511 question pairs), "Abdomen" (439 question pairs), and "Brain" (269 question pairs).

We also tested the method's performance when the number of clients spikes (500 clients) using the large-scale dataset PMC-VQA [45]. For the MRG task, we utilized the MIMIC-CXR-jpg [46] and IU X-Ray [47] datasets. The MIMIC-CXR-jpg dataset includes 370,000 medical images along with corresponding reports, for which we created 4 clients. The IU X-Ray dataset contains 7,470 medical images with associated reports, and we assigned 4 clients for this dataset. We adopted the original dataset partitioning, with client settings conducted separately on the training and test sets.

4.2. Evaluation Metrics

For the VQA task, we used answer accuracy as the evaluation metric. For VQG and MRG, we employed BLEU-4 [48], METEOR [49], ROUGE [50], and CIDEr [51] as evaluation criteria. All the scores of the language metrics were computed by the publicly available code provided by Chen et al. [52]. To mitigate the impact of accuracy fluctuations observed among some clients, attributed to the limited size of their test datasets, we determined the final accuracy based on the average results and the variance from the last round of local training.

Table 1: Quantitative results of accuracy comparison with previous medical VQA models. We report $Mean_{\pm Std}$ across 3 independent runs. The Enc. and Upd. denote the number of parameters in the baseline and parameters that need to be updated during training, respectively. We color code **best**, **second best** and **third best** performances.

Method	Slake $\uparrow$	VQA-RAD $\uparrow$	Enc. $\downarrow$	Upd. $\downarrow$
PMC-VQA [45]	78.4 $\pm$ 1.70	90.6 $\pm$ 0.82	7000	7000
M2I2 [53]	85.2 $\pm$ 0.95	72.1 $\pm$ 1.99	600	600
MEVF-BAN [54]	75.5 $\pm$ 1.45	75.7 $\pm$ 1.12	50	50
VGG+SAN [44]	76.1 $\pm$ 1.86	75.1 $\pm$ 0.59	50	50
FedAvg [4]	52.5 $\pm$ 2.43	46.7 $\pm$ 1.29	63	0.01
FedPer [55]	68.5 $\pm$ 1.34	65.7 $\pm$ 1.19	63	0.01
FedRep [56]	59.1 $\pm$ 1.67	59.1 $\pm$ 1.67	63	0.01
promptFL [57]	77.3 $\pm$ 1.13	73.3 $\pm$ 0.98	63	0.01
DSD-pFL [39]	79.2 $\pm$ 1.89	70.7 $\pm$ 1.63	63	0.3
DSD-pFL w prompt	63.9 $\pm$ 1.52	56.9 $\pm$ 0.60	63	0.01
DSD-pFL w prefix	78.2 $\pm$ 1.87	74.7 $\pm$ 1.29	63	0.01
DSD-pFL w LoRA	70.2 $\pm$ 0.38	68.2 $\pm$ 0.08	63	0.01

4.3. Experimental Setting

4.3.1. General Setting

We set the local epoch E_l to 10, the global epoch E_g to 8. The DLUC learning epoch for client communication E_d was set to 3, with an initial learning rate of 0.01, which was multiplied by 0.5 every 10 update steps. These values were empirically chosen to provide a good trade-off between local convergence, communication efficiency, and avoiding overfitting. The PEFT’s parameter settings were optimized experimentally. For prompt-tuning, we defined the prompt length as 6 for the shorter input sequences in VQG, whereas for the longer sequences in MRG, we set it to 20, as longer sequences generation requires a larger prompt capacity to capture complex contextual dependencies. The prompt was evenly divided into local and shared components. In prefix-tuning, learnable parameters were inserted in all 12 MHA layers, where the first 6 layers, learning more general information, were set as shared, while the latter 6 layers, focusing on local details, were designated as local. To balance parameter efficiency and adaptation expressivity, we set the rank of LoRA to 16 for the linear projection layers of all transformer-based models and introduced a dropout rate of 0.1. All clients share the encoders E_V , E_L and decoder M with a fixed weight, and only the clients’ PEFT parts are updated during training.

4.3.2. VQA Setting

We employed the pre-trained Contrastive Language–Image Pre-training (CLIP) [67] model as the image and text encoders E_V and E_L . The answer decoder M is implemented as a 2-layer MLP with a hidden size of 512 and a dropout rate of 0.2. The learning rate is initially 0.001, multiplied by 0.5 after every 20 steps, i.e., one complete local update. This initial learning rate was experimentally determined to provide a stable descent for the frozen CLIP and MLP parameters without diverging.

Comparison Methods Given the complexity and the substantial size of the baseline model employed, along with the intricacies of the multimodal task setup, direct comparison

with previous FL methods is not feasible. Considering that we aim to enhance the personalized performance of medical VQA, we compare the proposed with several state-of-the-art medical VQA models regarding accuracy and model parameters that need to be updated. Specifically, we employed PMC-VQA [45], M2I2 [53], MEVF-BAN [54], VGG+SAN [44], and DSD-pFL [39].

4.3.3. VQG Setting

The CLIP [67] model was used as the image encoder E_V , and the pre-trained GPT-2 [69] is used as the text encoder E_L and decoder M . The maximum length of the input sequence is set to 40. The maximum sequence length was constrained to 40 to accommodate the concise nature of task-specific questions while ensuring an optimal memory footprint. The learning rate is initially 0.001, multiplied by 0.5 after every 20 steps. Similar to VQA, this learning rate guarantees a stable adaptation for the PEFT modules.

Comparison Methods To verify our model’s effectiveness, we compared our model with other state-of-the-art VQG models. We first compared our model with the most original and simplest VQG model that uses a CNN and an RNN as a visual encoder and a language decoder, respectively. We also evaluated the performance of recently proposed models using the same VAE structure as ours, including medical VQG models: VQGR [58], MSVQG [61], and LMVQG [62] and two general VQG models: C3VQG [59] and Krishna et al. [60].

4.3.4. MRG Setting

We used CvT-21DistilGPT2 as the baseline method, which employed Convolutional vision Transformer (CvT) [70] pre-trained on ImageNet-21K as the image encoder E_V and a pre-trained DistilGPT2 as the decoder M . The maximum length of the input sequence is set to 128. A longer sequence length of 128 was chosen to accommodate the generation of complex, longer sentences typical in medical reports. The learning rate is initially $5e-5$, multiplied by 0.5 after every 20 steps. This learning rate was set smaller compared to VQA/VQG tasks to prevent catastrophic forgetting and maintain the stability of the deep hierarchical features inside the DistilGPT2 decoder during report generation.

Comparison Methods Apart from the baseline method, we selected three widely used methods for comparison: R2Gen [63], CMN [64], Contrastive Attention [65], and PPKED [66].

FL Baselines To validate our framework’s effectiveness against data heterogeneity and personalization needs, we benchmark against two representative methods: FedAvg [4] as the global baseline, and FedPer [55], which employs split-architecture strategies similar to our approach, serving as the primary pFL competitor.

4.4. Experimental Results

4.4.1. Comparison with Baseline Methods

As shown in Table 1, our approach achieves an average accuracy improvement of 3.5% over the baseline method

Table 2: Quantitative results of comparison with previous medical VQG models on Slake and VQA-RAD datasets. We report $Mean_{\pm Std}$ across 3 independent runs. We color code **best**, **second best** and **third best** performances.

Method	Slake				VQA-RAD			
	BLEU-4	METEOR	ROUGE	CIDEr	BLEU-4	METEOR	ROUGE	CIDEr
CNN+RNN	13.9 $\pm$ 1.61	14.6 $\pm$ 2.26	35.9 $\pm$ 0.90	57.8 $\pm$ 0.07	8.3 $\pm$ 1.93	12.0 $\pm$ 2.05	38.1 $\pm$ 2.11	52.9 $\pm$ 1.74
VQGR [58]	8.7 $\pm$ 1.04	14.9 $\pm$ 2.16	40.2 $\pm$ 2.26	63.5 $\pm$ 2.18	7.5 $\pm$ 0.11	11.8 $\pm$ 0.40	37.5 $\pm$ 1.96	63.9 $\pm$ 0.22
C3VQG [59]	4.6 $\pm$ 0.48	10.9 $\pm$ 2.32	27.8 $\pm$ 1.14	13.9 $\pm$ 2.02	4.3 $\pm$ 0.53	13.8 $\pm$ 0.78	29.1 $\pm$ 1.78	16.2 $\pm$ 1.67
Krishna [60]	10.9 $\pm$ 0.30	12.6 $\pm$ 1.38	40.9 $\pm$ 1.12	86.1 $\pm$ 0.17	13.2 $\pm$ 1.57	13.3 $\pm$ 0.17	38.3 $\pm$ 1.21	82.7 $\pm$ 1.46
MSVQG [61]	18.9 $\pm$ 1.47	18.0 $\pm$ 1.32	49.4 $\pm$ 1.75	77.0 $\pm$ 1.05	14.2 $\pm$ 0.21	22.0 $\pm$ 2.19	50.1 $\pm$ 1.92	77.5 $\pm$ 0.28
LMVQG [62]	26.8 $\pm$ 0.08	20.2 $\pm$ 1.87	44.9 $\pm$ 0.78	206.6 $\pm$ 1.41	29.9 $\pm$ 1.00	19.1 $\pm$ 1.09	44.0 $\pm$ 0.17	210.3 $\pm$ 1.29
FedAvg [4]	3.4 $\pm$ 1.45	8.1 $\pm$ 0.28	14.1 $\pm$ 2.17	6.9 $\pm$ 0.21	6.1 $\pm$ 1.48	10.8 $\pm$ 1.29	25.3 $\pm$ 1.84	9.6 $\pm$ 1.70
FedPer [55]	16.3 $\pm$ 0.82	16.4 $\pm$ 2.48	42.0 $\pm$ 0.61	67.0 $\pm$ 0.94	16.2 $\pm$ 0.16	18.8 $\pm$ 2.03	48.6 $\pm$ 1.10	72.3 $\pm$ 1.43
FedRep [56]	20.2 $\pm$ 1.25	18.1 $\pm$ 1.40	42.0 $\pm$ 1.18	147.1 $\pm$ 1.72	9.6 $\pm$ 0.86	12.9 $\pm$ 0.55	36.5 $\pm$ 1.04	73.4 $\pm$ 1.21
promptFL [57]	21.3 $\pm$ 1.96	20.1 $\pm$ 1.08	43.7 $\pm$ 0.91	158.9 $\pm$ 0.96	17.3 $\pm$ 1.87	16.9 $\pm$ 1.15	43.2 $\pm$ 1.71	147.4 $\pm$ 1.74
DSD-pFL w prompt	29.3 $\pm$ 0.73	23.1 $\pm$ 1.23	47.7 $\pm$ 1.75	206.4 $\pm$ 1.56	31.8 $\pm$ 1.13	29.7 $\pm$ 0.09	59.1 $\pm$ 1.05	319.9 $\pm$ 0.25
DSD-pFL w prefix	22.9 $\pm$ 0.69	20.8 $\pm$ 1.12	46.8 $\pm$ 0.10	159.0 $\pm$ 0.20	27.5 $\pm$ 1.31	27.7 $\pm$ 1.04	58.6 $\pm$ 1.18	285.6 $\pm$ 0.49
DSD-pFL w LoRA	20.4 $\pm$ 1.03	17.9 $\pm$ 0.99	41.7 $\pm$ 1.29	173.8 $\pm$ 0.90	24.6 $\pm$ 0.36	24.9 $\pm$ 0.43	55.8 $\pm$ 1.53	264.1 $\pm$ 0.49

Table 3: Quantitative results on IU X-Ray and MIMIC-CXR datasets. We report $Mean_{\pm Std}$ across 3 independent runs. We color code **best**, **second best** and **third best** performances.

Method	IU X-Ray				MIMIC-CXR			
	BLEU-4	METEOR	ROUGE	CIDEr	BLEU-4	METEOR	ROUGE	CIDEr
R2Gen [63]	10.5 $\pm$ 1.10	20.7 $\pm$ 0.27	39.2 $\pm$ 0.24	-	14.9 $\pm$ 1.44	10.8 $\pm$ 1.70	25.9 $\pm$ 1.24	-
CMN [64]	19.2 $\pm$ 0.93	19.4 $\pm$ 0.30	34.5 $\pm$ 1.04	-	9.9 $\pm$ 1.92	10.4 $\pm$ 2.03	29.5 $\pm$ 2.24	-
Cont [65]	14.4 $\pm$ 1.18	19.8 $\pm$ 0.10	37.5 $\pm$ 0.35	-	6.0 $\pm$ 1.75	14.5 $\pm$ 0.88	38.3 $\pm$ 0.13	-
PPKED [66]	11.8 $\pm$ 1.81	18.2 $\pm$ 1.75	-	35.4 $\pm$ 1.78	12.2 $\pm$ 0.38	15.7 $\pm$ 2.26	28.1 $\pm$ 1.25	-
CvT2 [65]	17.6 $\pm$ 1.59	18.7 $\pm$ 0.20	35.5 $\pm$ 2.27	54.5 $\pm$ 0.46	12.7 $\pm$ 2.23	16.6 $\pm$ 0.00	29.3 $\pm$ 1.71	37.4 $\pm$ 1.25
FedAvg [4]	15.5 $\pm$ 0.96	19.3 $\pm$ 1.56	34.6 $\pm$ 2.49	39.5 $\pm$ 0.23	12.3 $\pm$ 0.85	15.2 $\pm$ 0.35	26.7 $\pm$ 2.21	40.2 $\pm$ 0.39
FedPer [55]	16.1 $\pm$ 0.37	19.3 $\pm$ 2.31	35.5 $\pm$ 1.65	42.7 $\pm$ 1.28	12.4 $\pm$ 2.29	15.3 $\pm$ 0.24	27.7 $\pm$ 2.14	40.3 $\pm$ 0.61
FedRep [56]	12.5 $\pm$ 0.41	14.9 $\pm$ 1.76	28.1 $\pm$ 1.44	41.0 $\pm$ 1.10	15.7 $\pm$ 1.20	18.8 $\pm$ 0.61	36.1 $\pm$ 1.10	55.2 $\pm$ 0.49
promptFL [57]	13.2 $\pm$ 0.61	15.7 $\pm$ 0.77	29.0 $\pm$ 0.68	46.3 $\pm$ 0.42	16.1 $\pm$ 1.55	19.7 $\pm$ 1.48	39.9 $\pm$ 0.37	58.1 $\pm$ 1.92
DSD-pFL w prompt	17.1 $\pm$ 1.50	21.4 $\pm$ 0.61	39.5 $\pm$ 1.48	71.7 $\pm$ 0.77	12.0 $\pm$ 0.57	15.4 $\pm$ 1.67	27.4 $\pm$ 1.36	39.5 $\pm$ 0.71
DSD-pFL w prefix	15.8 $\pm$ 0.37	18.4 $\pm$ 1.71	36.1 $\pm$ 0.36	39.3 $\pm$ 1.92	9.0 $\pm$ 0.76	12.2 $\pm$ 1.76	27.1 $\pm$ 1.87	16.4 $\pm$ 0.07
DSD-pFL w LoRA	16.0 $\pm$ 0.36	17.9 $\pm$ 0.14	37.1 $\pm$ 0.21	47.7 $\pm$ 0.31	10.4 $\pm$ 1.58	11.5 $\pm$ 1.39	25.9 $\pm$ 0.54	8.6 $\pm$ 1.94

DSD-pFL [39] under the same task setup, demonstrating its effectiveness in pFL for medical VQA. Compared to locally trained models with similar baseline parameter scales, our method—particularly with prefix-tuning—achieves superior performance, though a performance gap remains when compared to methods utilizing significantly larger parameter baselines. Notably, our method requires updating only 0.01% of the parameters in Vision Transformer-based models and merely 0.00001% of the parameters in current state-of-the-art models, significantly reducing computational and communication costs. Furthermore, a notable discrepancy in prefix-tuning outperforms prompt-tuning in VQA. This divergence is attributed to the nature of task-specific information processing. In VQA, prefix-tuning enhances the self-attention mechanism within MHA layers, allowing for more precise cross-modal feature alignment essential for answer prediction.

In the VQG and MRG tasks, as shown in Tables 2 and 3, our proposed DSD-pFL achieves higher scores in most metrics when selecting the appropriate PEFT method (Due to the large number of evaluation metrics for generating tasks and space constraints, we omitted the variance from the table). Notably,

prompt-tuning yields the best results in both tasks, suggesting that lightweight modifications to input embeddings are particularly effective for medical vision-language generation. This advantage arises from its ability to provide a flexible guiding mechanism. For VQG, effective question generation requires a controlled fusion of image and textual cues, where prompt-tuning acts as an implicit guide, refining the model’s understanding of relevant attributes. In MRG, where reports demand contextual depth and coherence, prompts help steer the generation process toward clinically meaningful descriptions. In contrast, Prefix-Tuning and LoRA primarily adjust internal model representations rather than providing explicit task guidance. This makes them less effective in VQG and MRG, where structured text generation benefits more from direct input-level modifications that steer the model’s output in a task-specific manner.

Notably, in the MIMIC-CXR MRG task, baseline personalization methods (promptFL and FedRep) outperformed our proposed DSD-pFL variants. This is primarily attributed to the characteristics of the MIMIC-CXR dataset: consisting of 370,000 images, it is massive but relatively homogeneous compared to multi-institutional

Table 4: Comparison of VQA accuracy (%) across several common baseline networks before and after applying the proposed method.

Baseline	Param.	Slake				VQA-RAD		
		Client1	Client2	Client3	Client4	Client1	Client2	Client3
VIT-B/32 [67]	63M	-1.2 $\pm$ 1.0	+1.8 $\pm$ 0.8	+0.9 $\pm$ 0.2	+4.7 $\pm$ 1.7	+2.8 $\pm$ 1.6	+2.1 $\pm$ 0.1	+0.6 $\pm$ 0.2
PubMedCLIP [68]	63M	-0.3 $\pm$ 0.5	+0.3 $\pm$ 0.9	+6.8 $\pm$ 0.9	+4.7 $\pm$ 0.4	+0.2 $\pm$ 1.2	-2.3 $\pm$ 1.4	+2.7 $\pm$ 1
VIT-L/14 [67]	300M	-0.7 $\pm$ 1.0	+0.3 $\pm$ 1.2	+0.2 $\pm$ 0.8	+3.0 $\pm$ 0.7	+2.8 $\pm$ 2.1	+2.4 $\pm$ 1.4	-1.52 $\pm$ 2.5
ViT-L/14@336px [67]	300M	-1.2 $\pm$ 0.2	-2.7 $\pm$ 0.6	+2.3 $\pm$ 0.4	+9.5 $\pm$ 2.9	+1.1 $\pm$ 0.8	+2.0 $\pm$ 1.4	-0.4 $\pm$ 2.5

Table 5: Ablation study on SLAKE and VQA-RAD using DSD-pFL with prefix-tuning. This table evaluates the impact of FL, DLUC, and brief-based uncertainty weighting under CLOSED and OPEN settings. Bolded values indicate the best results for each client.

Dataset	ID	Client	FL	DLUC	Brief	CLOSED				OPEN			
						Client1	Client2	Client3	Client4	Client1	Client2	Client3	Client4
SLAKE	(a)					77.8 $\pm$ 0.2				58.2 $\pm$ 0.4			
	(b)	✓				83.7 $\pm$ 0.4	71.3 $\pm$ 0.7	80.8 $\pm$ 0.6	74.9 $\pm$ 1.6	59.5 $\pm$ 0.2	56.3 $\pm$ 0.8	51.1 $\pm$ 0.9	43.8 $\pm$ 0.8
	(c)	✓	✓		✓	79.9 $\pm$ 0.4	74.8 $\pm$ 0.6	80.9 $\pm$ 0.4	78.4 $\pm$ 0.4	65.9 $\pm$ 0.1	55.9 $\pm$ 0.2	63.7 $\pm$ 0.5	61.1 $\pm$ 0.8
	(d)	✓	✓	✓		78.2 $\pm$ 0.3	63.9 $\pm$ 0.5	80.5 $\pm$ 0.2	69.2 $\pm$ 0.2	62.3 $\pm$ 0.4	51.3 $\pm$ 0.6	62.4 $\pm$ 0.5	58.8 $\pm$ 0.7
	(e)	✓	✓	✓	✓	81.7 $\pm$ 0.5	74.7 $\pm$ 0.8	81.3 $\pm$ 0.3	78.9 $\pm$ 1.0	66.0 $\pm$ 0.3	56.1 $\pm$ 0.3	63.8 $\pm$ 0.2	61.7 $\pm$ 0.8
VQA-RAD	(a)					67.5 $\pm$ 0.9				42.6 $\pm$ 0.4			
	(b)	✓				68.8 $\pm$ 0.7	62.9 $\pm$ 1.6	80.8 $\pm$ 1.4	-	44.2 $\pm$ 0.2	38.2 $\pm$ 0.3	34.5 $\pm$ 0.1	-
	(c)	✓	✓		✓	71.0 $\pm$ 0.5	61.8 $\pm$ 0.6	76.5 $\pm$ 0.8	-	47.7 $\pm$ 0.2	39.9 $\pm$ 0.7	35.3 $\pm$ 0.2	-
	(d)	✓	✓	✓		63.1 $\pm$ 1.1	60.2 $\pm$ 0.2	74.8 $\pm$ 1.5	-	43.8 $\pm$ 0.4	37.4 $\pm$ 0.8	30.7 $\pm$ 0.3	-
	(e)	✓	✓	✓	✓	72.1 $\pm$ 1.2	65.1 $\pm$ 0.9	81.2 $\pm$ 0.8	-	47.9 $\pm$ 0.2	40.3 $\pm$ 0.3	34.3 $\pm$ 0.3	-

datasets that exhibit extreme heterogeneity. Because promptFL uses a unified prompt without decoupling global and local representations, it maintains a larger effective capacity for shared feature-space optimization. This allows it to aggregate generic medical knowledge more effectively when subspace mismatches are less severe. Conversely, while DSD-pFL is intentionally engineered to prevent negative transfer under highly non-IID conditions, in homogeneous settings this decoupling acts as a restrictive capacity bottleneck, decreasing performance. This highlights an important architectural trade-off: DSD-pFL excels when client distributions exhibit significant divergence, whereas unified PEFT mechanisms like promptFL may hold an edge in large-scale, low-heterogeneity data silos. To demonstrate that our methodology can be effectively extended to all transformer structure networks, we tested the performance of our method on multiple baselines. As shown in Table 4, our method can effectively improve the personalization performance of multimodal transformers, and we achieved an *average of 3% improvement* on most clients.

Our experimental results strongly support the effectiveness of our proposed approach, which can be attributed to two key factors. First, PEFT mitigates overfitting to site-specific biases in medical data distributions while preserving the model's capacity to encode critical domain knowledge. This selective fine-tuning approach ensures that personalized adaptations remain clinically meaningful, preventing degradation of general medical reasoning while allowing client models to capture institution-specific variations in imaging protocols and diagnostic language. Moreover, DLUC dynamically adjusts parameter aggregation weights based on the uncertainty. This allows the framework to down-weight unreliable updates from

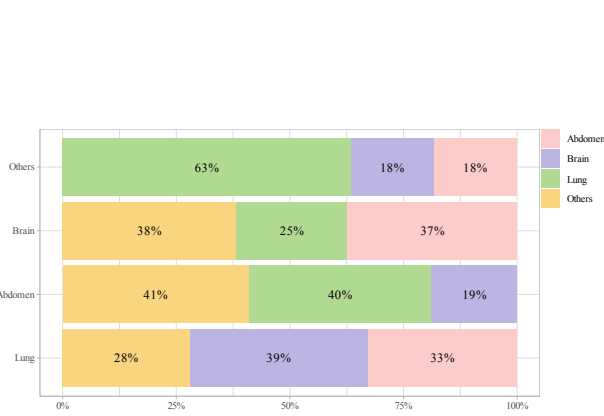
clients whose local data distributions exhibit high variance or limited representativeness in the global context. By incorporating DST-based belief fusion, DLUC prevents erroneous parameter shifts that exacerbate client drift, ensuring that the aggregated model maintains robust generalization across heterogeneous medical datasets while allowing local models to refine task-specific feature representations within their respective data domains.

We took the scenario of prefix-tuning in the VQA task as an example for the following analysis. Similar experiments were conducted on other tasks and with other PEFT methods, yielding the same conclusions.

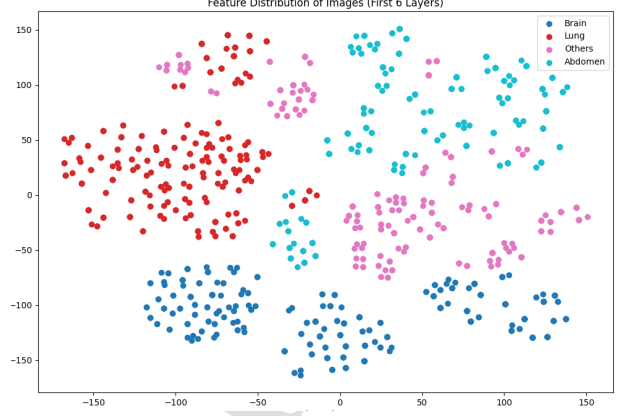
In addition to standard comparisons, we specifically evaluate the advantages of our framework against FL baselines. As presented in Table 1, the global baseline FedAvg exhibits poor performance, particularly on VQA-RAD with an accuracy of 46.7%, highlighting the "Negative Transfer" phenomenon caused by the severe non-IID distribution of medical data. While the basic personalized method FedPer improves performance by splitting the model into shared and local components, it still lags behind our approach. Our method outperforms FedPer by approximately 10% on Slake and 5% on VQA-RAD. This substantial margin confirms that simply splitting the model is insufficient. Instead, dealing with the high-dimensional manifold of medical VLMs requires a more nuanced strategy like our DLUC-guided PEFT, which ensures valid knowledge sharing while preventing subspace mismatch.

4.4.2. Ablation studies

We designed: trained on the whole data without setting up the client, setting up the client but without parameter sharing in



(a) Inter-client communication weights assigned by DLUC. The stacked bars indicate the proportion of knowledge each client incorporates from other clients, demonstrating the adaptive weight allocation in federated learning.



(b) Feature distribution of images in the SLAKE dataset extracted from the first six layers of CLIP. Different colors represent distinct anatomical regions, illustrating the inherent feature clustering among different clients.

Figure 2: (a) Adaptive inter-client communication weights and (b) feature distribution among clients in the SLAKE dataset.

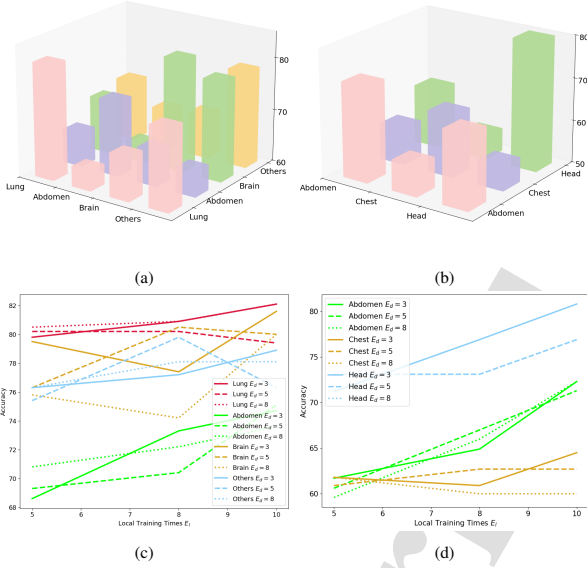


Figure 3: (a,b) Cross-client evaluation results showing a significant drop in accuracy when prompts are exchanged, indicating that prompts capture client-specific information rather than generalizable features. (c,d) Results of testing different settings of E_l with different E_d under the SLAKE and VQA-RAD datasets.

server, without DLUC for information aggregation (equal weights, i.e. FedAvg), and without brief that used the randomly generated brief value instead of the result calculated by DST. As shown in Table 5, the results demonstrate that DLUC can effectively aggregate clients' information to improve the performance of each client, and the brief generated by DST effectively represents client performance and provides guidance for the generation of final weights. Notably, the performance gains from DLUC and DST-based

briefing vary across clients, which is attributed to the differing degrees of local data heterogeneity and distribution shift; clients with more severe subspace mismatch benefit more significantly from the uncertainty-aware filtering. We conducted similar experiments on other tasks and with other PEFT methods, and obtained similar results.

4.4.3. Complexity and memory

For T participating clients, b the bytes per scalar, S the size of the shared PEFT block, and S_{full} the full-model size with $S \ll S_{\text{full}}$. The shared size is $S_{\text{prompt}} = dm$ for prompt-tuning, $S_{\text{prefix}} = L_s dm$ for prefix-tuning, and $S_{\text{LoRA}} \approx L_s q (2mr)$ for LoRA, where d is prompt length, m hidden size, L_s the number of shared layers, q the number of adapted projections, and r the LoRA rank. In each global round, every client uploads and downloads $O(S)$ scalars, so the total traffic is $2bKS$ bytes; the server performs a weighted sum in $O(KS)$ time and, if a light reweighting runs E_d iterations on client-wise scalars, this adds $O(E_d K)$, hence server time is $O(KS + E_d K)$ and typically dominated by $O(KS)$ when $S \gg E_d$. Compared to FedAvg, which communicates $2bKS_{\text{full}}$ bytes and aggregates in $O(KS_{\text{full}})$ time, the round-wise ratios obey:

$$R_{\text{traffic}} = \frac{2bKS}{2bKS_{\text{full}}} = \frac{S}{S_{\text{full}}} \ll 1,$$

$$R_{\text{time}} = \frac{O(KS + E_d K)}{O(KS_{\text{full}})} \approx \frac{S}{S_{\text{full}}} \quad (\text{when } S \gg E_d),$$

and the same factor applies to memory. Plugging representative values $m=768$, $d=20$, $L_s=6$, $q=4$, $r=16$ (FP32) gives $S_{\text{prompt}}=15,360$ (≈ 60 KB), $S_{\text{prefix}}=92,160$ (≈ 0.35 MB), and $S_{\text{LoRA}} \approx 589,824$ (≈ 2.3 MB). Taking a typical backbone size $S_{\text{full}} \approx 10^8$ scalars, the reduction factors are $S_{\text{full}}/S_{\text{prompt}} \approx 6.5 \times 10^3$, $S_{\text{full}}/S_{\text{prefix}} \approx 1.1 \times 10^3$, and $S_{\text{full}}/S_{\text{LoRA}} \approx 1.7 \times 10^2$, i.e., our communication, aggregation time, and memory are smaller by these multiplicative factors, with identical linear scaling in K . Quantization to q bits simply replaces b by $q/8$ and preserves all ratios.

4.4.4. Analysis of inter-client weights

Fig. 2a illustrates the weights generated by DLUC during inter-client communication, revealing significant differences in dependencies among clients. For instance, since the ‘Other’ dataset includes lung images as well, it explains why the ‘Other’ client assigns the highest weight (63%) to the lung client. Additionally, due to the widespread use of MRI in brain and abdominal imaging and their similar imaging characteristics, such as soft tissue contrast, the brain client assigns relatively high weights to the “abdomen” (37%) and “Other” (38%) clients. From a feature distribution perspective, Fig 2b further validates these findings by showing the feature distributions extracted from the first six layers of CLIP for images in the SLAKE dataset. The visualization highlights that different organ datasets exhibit clear distribution differences, supporting the notion that each client’s data has distinct feature representations. However, certain degrees of overlap, particularly between the “Other” and “lung” datasets, suggest that some clients share similarities in their imaging characteristics. This distribution pattern explains why DLUC assigns varying weights to different clients, effectively facilitating personalized inter-client knowledge transfer in FL.

4.4.5. Cross evaluation between prompts

To further validate that prompts effectively personalize the model rather than merely capturing generalizable information, we conducted a cross-client evaluation, where prompts trained on one client were applied to another client’s data. As shown in Fig. 3a and 3b, there is a significant drop in accuracy when prompts are exchanged, indicating that prompts primarily learn client-specific data distributions rather than general features. This result reinforces the effectiveness of our pFL approach, demonstrating that it successfully adapts to the unique characteristics of each client’s data rather than converging towards a generic global representation, thereby enhancing model personalization.

4.4.6. Impact of hyperparameters

The model’s performance is affected by two primary parameters: the local training step E_l , the DLUC step in inter-client communication E_d . The comparison results on steps are shown in Fig. 3c and 3d. Excessive local training steps can lead to over-localization of clients, making it difficult to aggregate due to significant differences between clients. On the other hand, too few local training steps may result in client models not adequately learning the local dataset. Furthermore, overly large or small DLUC steps will both lead to failure in learning the weights.

4.4.7. Scalability Stress Test with 500 Clients

To evaluate the framework’s scalability and robustness under extreme concurrency, we conducted a stress test simulating 500 participating clients using the PMC-VQA dataset. In this scenario, clinical data was randomly partitioned across clients to simulate a large-scale, high-density hospital network focused on throughput rather than extreme heterogeneity.

Acknowledging that random partitioning yields an IID environment, this experiment specifically assesses system orchestration and VRAM footprint rather than non-IID subspace mismatch. Thanks to the ultra-lightweight nature of PEFT (only ~0.01M trainable parameters per client), our framework successfully managed this massive federation consuming only 13GB of VRAM. Results showed stable convergence with an average 1.8% accuracy improvement across 56% of clients, confirming that DLUC maintains aggregation stability even when the number of peers scales by two orders of magnitude.

5. Conclusion and Discussion

In this study, we proposed a reliable pFL framework for vision-language medical applications, specifically targeting tasks such as medical VQA, VQG, and MRG. By leveraging PEFT methods integrated within multi-head attention layers, our approach significantly reduced computational overhead while maintaining personalization capabilities. Furthermore, we introduced a DLUC process, based on Dempster-Shafer Theory, to optimize the aggregation of client-specific updates and mitigate client drift. Extensive experiments demonstrate the superior performance of our method compared to existing FL approaches, particularly in terms of aggregating meaningful information while preserving client-specific model adaptation. The results confirmed that our pFL method effectively balances generalization and personalization, offering an efficient solution to the challenges of data heterogeneity and privacy in medical AI. Notably, this work represents the first attempt to apply pFL in training personalized multimodal medical models on non-IID data. The integration of PEFT and uncertainty calibration opens new pathways for developing reliable and scalable medical AI under federated settings, advancing the field of personalized medical AI.

Ultimately, the clinical utility of our framework lies in its ability to deploy Expert-in-the-Loop AI systems without centralized data aggregation. By allowing hospitals to retain their unique data characteristics through personalized subspaces while benefiting from global medical knowledge leveraging shared subspaces, our method fits seamlessly into privacy-sensitive clinical workflows. This strategy effectively reduces the cold start problem for new hospitals and ensures that AI diagnostic suggestions remain relevant to local demographic and device specifics, fostering greater trust and adoption among clinicians. Future work will focus on integrating more adaptive parameter-sharing strategies and exploring privacy-preserving techniques such as Differential Privacy to further harden the security of the shared manifolds. A limitation of the current study is that the evaluation is conducted on public datasets with simulated client partitions rather than large-scale, real-world multicenter cohorts, such as the UK Biobank or eICU. Validating our framework on such extensive, multi-institutional datasets remains a critical direction for future work.

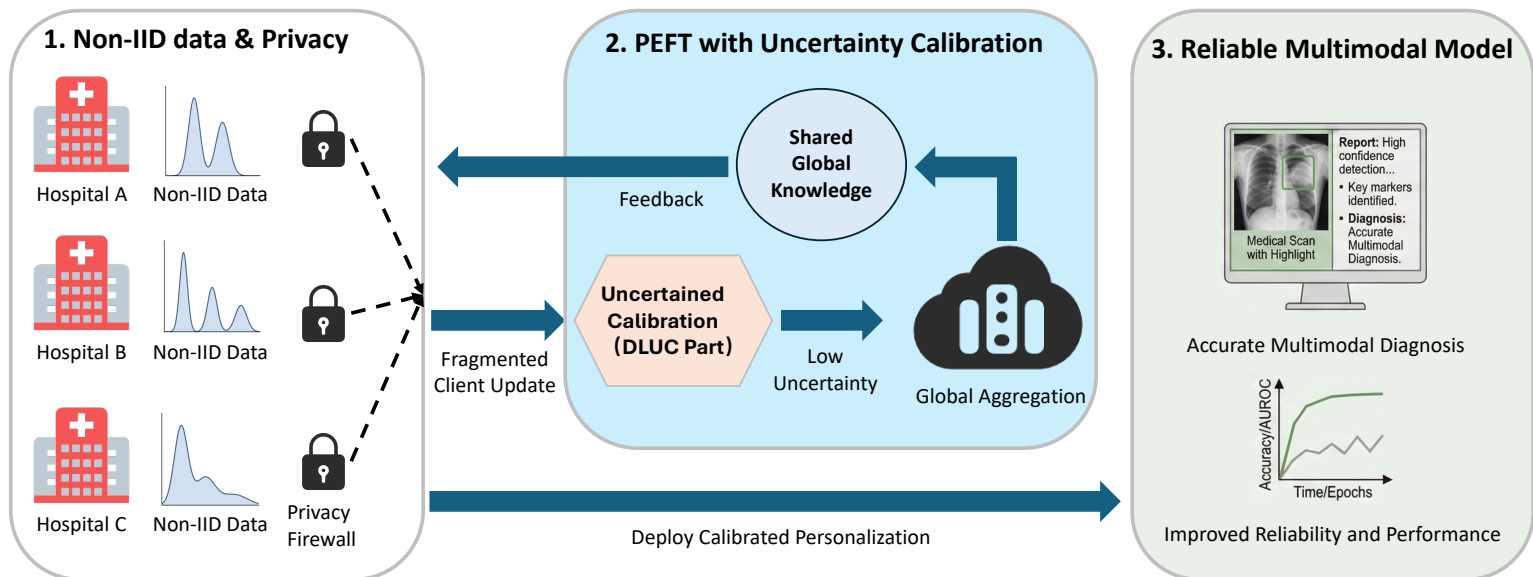
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